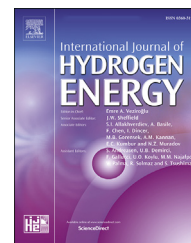


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Copper-based catalysts for ethanol dehydrogenation and dehydrogenative coupling into hydrogen, acetaldehyde and ethyl acetate

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HIGHLIGHTS

- The progress of ethanol dehydrogenation over Cu-based catalysts is summarized.
- The selectivity to acetaldehyde and ethyl acetate can be tailored.
- Particle size (<10 nm) and dispersion (20–60%) of Cu is suitable.
- Cu/MgAlO_x, Cu/ZrO₂ and Cu/SiO₂/SiC are potential catalyst candidates.

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ABSTRACT

Hydrogen is clean energy for the future and bio-hydrogen can be produced from bio-ethanol by dehydrogenation, and subsequent dehydrogenative coupling reactions. Two main products, acetaldehyde, and ethyl acetate, along with hydrogen are important industrial products from ethanol. In this work, we focus on the progress on the development of Cu-based catalysts for ethanol dehydrogenation and dehydrogenative coupling reactions. The review discusses the effect of Cu charge, Cu particle size and dispersion, support, promoters and reaction conditions. This work found that the rate of ethanol consumption is different from the absolute ethanol conversion. Cu loading and Cu dispersion have a strong impact on the catalytic activity and the proper value are 10% and 20–60% for Cu loading and dispersion, respectively. The selectivity to acetaldehyde and ethyl acetate can be tailored by particle size, Cu oxidation, acidity/basicity of catalyst as well as contact time. This review will provide a full picture of the development of Cu-based catalysts for ethanol dehydration and has a significant impact on the field.

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Introduction

Hydrogen is potentially clean energy and a zero air pollution due to the formation of water as a product after combustion, making a decrease in greenhouse gas emissions [1–4].

Hydrogen is high-efficient energy (120 MJ/kg) in comparison with liquid hydrocarbons-based fuels and is widely used in the fuel cell system to generate electricity [1–3,5]. Additionally, hydrogen has wide applications in the industry including consumption in processes (e.g. ammonia production, upgrading heavy oils), coolant, energy carrier, and semiconductor

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industry [6–11]. Currently, hydrogen is mostly produced from the reforming process of fossil hydrocarbon [12–15], including 48% from natural gas, 30% of heavy oils and naphtha, and 18% from coal, and the rest from water electrolysis [16]. Additionally, hydrogen can be also produced by thermochemical hydrogen production processes, mainly including thermochemical water-splitting cycles [17,18] and thermochemical biomass conversion [19]. In thermochemical water-splitting, the sulfur-iodine (S–I) and copper-chlorine (Cu–Cl) cycles are the significant global interests due to the lowest cost (US \$1.77/kg H₂ for 4-step Cu–Cl), highest energy efficiency (57% for 3-step Cu–Cl), and highest exergy efficiency (78.21% for open-loop Zn–S–I) [17,18]. Besides that, the combination of electrolysis and thermochemical water-splitting can avoid the drawback of separate techniques resulting in a cost-efficient water-splitting process [20]. Hydrogen can be produced from biomass by thermochemical techniques, and steam reforming, pyrolysis, and in-line reforming are the high-efficiency processes for hydrogen production from biomass [19].

Moreover, hydrogen can be produced from ethanol [21–28], a green material through the pretreatment and fermentation process of lignocellulosic [29], date wastes [30], and sugarcane [31], and globally growing up to 41.4 billion liters by 2025 [32]. The ethanol production from lignocellulosic biomass meets a significant challenge in the pretreatment of materials [29,30], while ethanol production from sugarcane still has a challenge in a better understanding of the changes of the microbiological state including the colonization of yeasts in the tanks, the knowledge of the most prevalent yeasts, the stress of yeasts, the isolation of yeasts resistance, the understanding of the different metabolic pathways, and the genomic constitution of stress-resistant yeasts [31]. However, with the development of new techniques in pretreatment and knowledge of yeasts, as well as the low price of date wastes, the production of ethanol will increase and the price will decrease [29–31]. Therefore, hydrogen production from green bioethanol is excellent energy for the future. Hydrogen has been currently made from ethanol through ethanol steam reforming and ethanol dehydrogenation. Ethanol steam reforming has an advantage in the lowest cost of manufacturing (\$1679.96/ton of H₂), fixed capital investment (\$1,012,435.06/1693.04 ton of H₂/year) and H₂ production cost (\$1.58/kg H₂) in comparison to ethanol dehydrogenation to ethyl acetate process (\$3298.32/ton of H₂; \$4,784,140.76/1664.77 ton of H₂/year; \$3.24/kg H₂) and ethanol dehydrogenation to acetaldehyde process (\$3005.21/ton of H₂; \$1,577,309.15/1663.85 ton of H₂/year; \$1.97/kg H₂) [25]. However, the ethanol steam reforming process releases a high amount of CO₂ emission (10.21 kg CO₂/kg H₂) than that of ethanol dehydrogenation to ethyl acetate process (0.80 kg CO₂/kg H₂) and ethanol dehydrogenation to acetaldehyde process (2.94 kg CO₂/kg H₂) [25]. Therefore, the dehydrogenation of ethanol (CH₃CH₂OH → CH₃CHO + H₂) [33–37], and coupling reaction of acetaldehyde with ethanol to form ethyl acetate ester (C₂H₅OH + CH₃CHO → CH₃COOC₂H₅ + H₂) [38] are the potential processes. The second reaction enhances the yield of hydrogen from ethanol, instead of dehydrogenation of ethanol only.

Besides hydrogen, acetaldehyde and ethyl acetate are also products from the dehydrogenation and coupling reaction of

ethanol. Currently, acetaldehyde is mostly produced from ethylene oxidation (Hoechst-Wacker process). It has a wide application in the industry such as food industry, plastic industry, the manufacture of acetic acid, paint binders in alkyd paints, and is used as a component or in the production of materials in civil/pharmaceutical/cosmetics industries [39–43]. The market of acetaldehyde was US\$1.26 Billion in 2016 [44] and reaches 1.6 million tons by 2024 [45] and US\$2.1 billion by 2027 [46] (Fig. 1). In 2020, the global market for acetaldehyde is estimated at around US\$1.4 billion due to the Covid-19 crisis even it was forecasted to reach US\$2 Billion [46]. For ethyl acetate, it is an important industrial solvent using mostly in paints & coatings, printing ink, process solvents, food & beverage, with a global market of around US\$3.3 Billion in 2019 and is a project to reach US\$4.4 Billion by 2027 [47] (Fig. 1). Both global market value of acetaldehyde and ethyl acetate is dramatically lower than that of hydrogen (Fig. 1) due to the high demand for hydrogen for many industries.

Therefore, the production of hydrogen, acetaldehyde, and ethyl acetate is very essential. In which, the production of those materials from “green” ethanol is necessary to reduce the use of fossil resources and enhance the quality of the environment. Homogeneous catalysts, such as [RuH₂(-CO)(PPh₃)₃], shows high ethanol conversion (>90%) at mild condition (~100 °C) [48]. However, separation of homogeneous is a significant challenge in the catalytic process, therefore, the heterogeneous catalysts have received more attention for the ethanol conversion process. There are many heterogeneous catalysts used for the dehydrogenation of ethanol into acetaldehyde and ethyl acetate such as Au [49], NiAu [50], AuPd [51,52], Pd [53], PdZn [54], Co [22], AgLi [55], Ag-Ni [56], CeO_x [57], mesocellular form carbon [58], and Cu [38,59–61]. Among them, copper is considered one of the best metal catalysts for ethanol dehydrogenation due to its low price and high activity. In fact, copper doped on the active carbon showed 65.3% ethanol conversion with 96.3% acetaldehyde selectivity at 350 °C higher than that of Ni, Co, Ce doped on the active carbon (Fig. 2) [62]. Recently, copper-based catalysts are applied for the production of ethyl acetate from ethanol directly in both laboratory [63] and pilot [64] scales. In the pilot-scale study, selectivity to ethyl acetate obtained is around 89 to 75% with ethanol conversion of 31–58% at 230–270 °C. Interestingly, the copper catalysts here can remain the activity without any loss for 1250 h [64]. Therefore, copper is one of the potential active metals for ethanol dehydrogenation into hydrogen and acetaldehyde, and further coupling into ethyl acetate.

For the review of this field, Sun et al. [65] and Nada et al. [66] briefly mentioned the catalysts and reaction mechanism for ethanol dehydrogenation reaction. While, Giannotti et al. [67] focused on the catalytic process of hydrogen production via dehydrogenation of organic compounds, including ethanol. Recently, Huang et al. has published a mini-review summarizing the use of heterogeneous catalysts in ethanol dehydrogenation including noble metal catalysts, non-metal catalysts, alloyed nanocatalysts, nitride and carbide catalysts, and metal oxide catalysts [68]. In which, Cu and Co catalysts are the main catalysts used in non-metal catalysts for this reaction. The

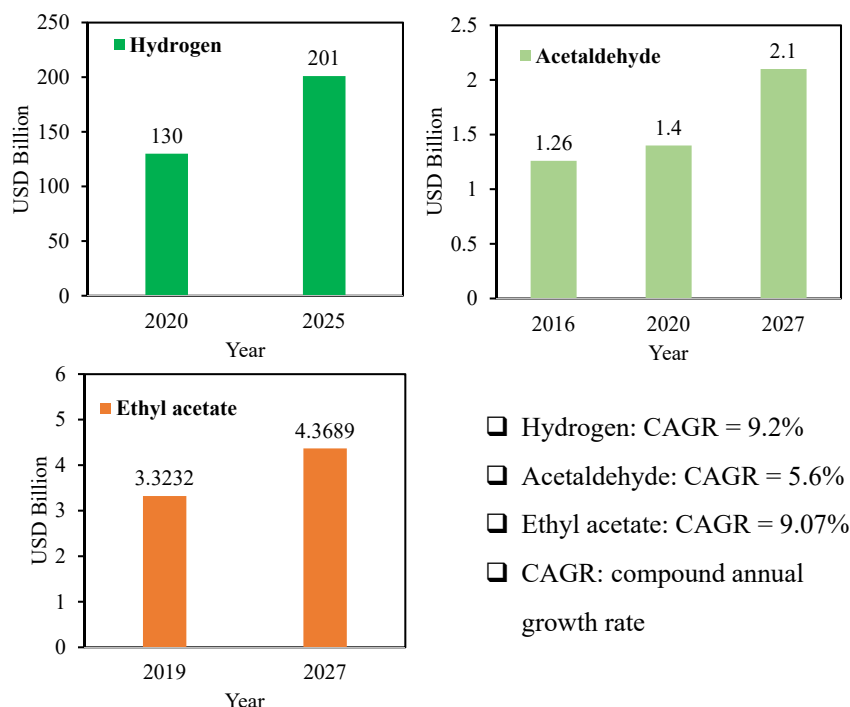


Fig. 1 – The global market value of hydrogen (reproduced from Ref. [96]), acetaldehyde (reproduced from Refs. [44,46]), and ethyl acetate (reproduced from Ref. [47]).

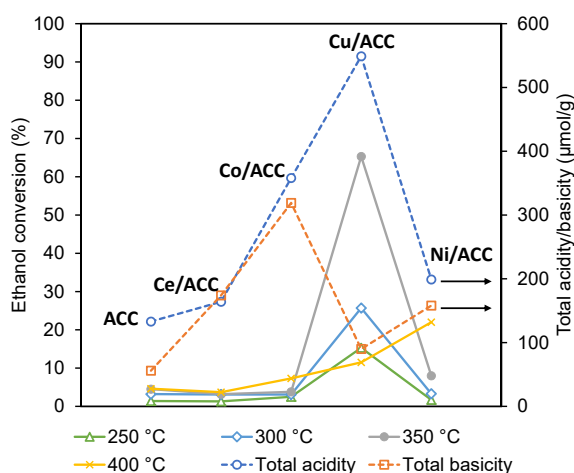
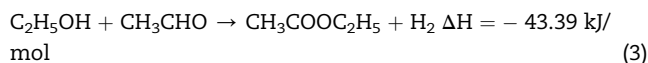
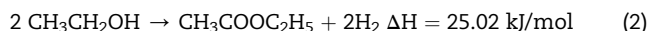


Fig. 2 – Ethanol conversion at different temperature and total acidity/basicity of Cu-based catalyst. Reproduced from Ref. [62].

review summarized the progress of the catalysts for ethanol dehydrogenation. In this work, we focus on only copper-based catalysts and summarize the recent progress in ethanol dehydrogenation and a secondary coupling of acetaldehyde into hydrogen over copper-based catalyst. We will deeply discuss the parameters including copper metal, support, the metal addition and reaction condition, affecting ethanol dehydrogenation and coupling reaction as well as reaction mechanism and the stability of catalysts to elucidate the current development of this field.

Reaction mechanism

The reaction mechanism of ethanol dehydration into acetaldehyde and hydrogen over Cu-based catalysts has widely been investigated [38,62,69–72]. Looking at the chemical reactions to form hydrogen, acetaldehyde, and ethyl acetate from ethanol:



Reaction (1) and (2) are favored thermodynamically at high temperatures, and reaction (3) is favored thermodynamically at low temperatures. Reaction (3) of ethyl acetate production is exothermic because it is a consequence of reaction (1) and (2) with acetaldehyde production as intermediate step [73]. By saving energy, ethanol can transform to ethyl acetate at low temperature via acetaldehyde as an intermediate.

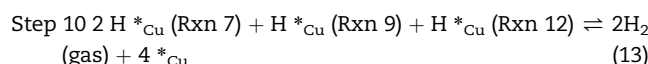
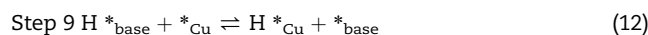
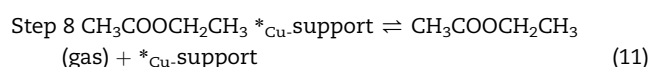
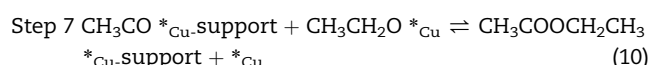
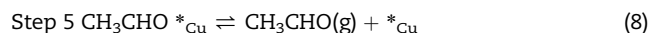
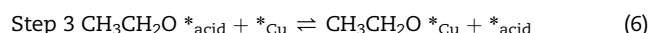
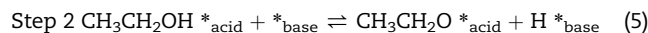
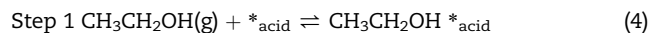
According to the literature, ethanol dehydrogenation occurs on the surface of Cu, ethanol dehydration occurs on the surface of a support, and dehydrogenative coupling reaction replies on the interface of copper-support [38,69–71]. Based on the current observation of Finger et al. [38], the dehydrogenation of ethanol happens on the surface of Cu and the dehydrogenative coupling occurs at the interface of copper-metal oxides, i.e. ZnO and ZrO₂, but not on strong Lewis

acid Al_2O_3 (Fig. 3). Pang et al. [74] found that acetaldehyde is generated on the surface of Cu^+ species, while H_2 is liberated on the surface of Cu^0 species. In addition, the support is also important because turnover frequency (TOF, the number of molecules converted per active site and per unit of time) of acetaldehyde and ethyl acetate are dependent on the enthalpy of adsorption of ethanol on the surface of metal oxide support [38]. Suggesting that surface of Cu particles, support and the interface of copper-metal oxide can play an important role in the reaction rate of acetaldehyde and ethyl acetate.

Moreover, both computational and experimental results reported that the rate-determining step is the ethanol activation into ethoxy species on Cu-based catalysts [69,70]. Those ethoxy species adsorbed the surface of the support, therefore, support is very important in the ethanol dehydrogenation and dehydrogenative coupling over Cu-base catalysts. Recently, Hanukovich et al. [72] reported that O–H cleavage is a rate-limiting step over Cu-based catalysts at low ethanol partial pressure (~1 mbar), while the rate-limiting step switches to $\text{C}_\alpha\text{--H}$ cleavage at $\text{Cu}^{\delta+}$ surface at high ethanol partial pressure (~70 mbar). The C–H bond cleavage is the rate-determining step in ethanol dehydrogenation reaction also performed by Shan et al. [75]. Proposing that both support and Cu have important roles and act cooperatively to this reaction as well as the reaction condition, e.g., ethanol pressure, has a strong effect on the rate-determining step of this reaction.

For the detail of mechanism, Hanukovich et al. [72] proposed the reaction mechanism of ethanol dehydrogenation over Cu/support and combine with the mechanism of ethyl acetate formation from ethanol reported by Colley et al. [69] and Wu et al. [63]. Additionally, ethyl acetate is found on the interface of copper-support [38]. We propose the reaction mechanism of ethanol dehydrogenation and dehydrogenative

coupling reaction as following steps, in which $^*\text{Cu}$ represents Cu adsorption sites, $^*\text{acid}$ represents acid sites, and $^*\text{base}$ represents base sites.



In this mechanism, ethanol is adsorbed to an acid site of support (Step 1), then cleavage O–H bond to form ethoxy species on acid support and H radical on basic support (Step 2). The ethoxy species on the acid site diffuses to an interfacial Cu site (Step 3) followed by $\text{C}_\alpha\text{--H}$ bond cleavage (Step 4). Next, acetaldehyde on interfacial Cu site desorbed to form acetaldehyde (Step 5) or cleave $\text{C}_\alpha\text{--H}$ bond to form H radical on an interfacial Cu site and acetyl species on an interfacial Cu-support site (Step 6). Then, acetyl and ethoxy species from ethanol adsorption react to form ethyl acetate species on an interfacial Cu-support site (Step 7) followed by desorption to form ethyl acetate (Step 8). In Steps 9 and 10, the H atom formed on the base site in Step 2 diffuses to the Cu site and is desorbed from the Cu surface to form H_2 .

Copper metal

Based on the mechanism of this reaction, copper metal is the most important factor relative to the activity and product's selectivity. The Cu charge, Cu loading, particle size, Cu dispersion, the interface of Cu-support can tailor the catalytic activity of Cu [60,76].

The charge of Cu particles

The charge of Cu particles strongly impacts the product's selectivity in ethanol dehydrogenation [60,77]. The formation of acetaldehyde and ethyl acetate depends on the electronic properties of Cu particles, i.e. the ratio of Cu^0/Cu^+ [60]. Indeed, observed that the highest selectivity to ethyl acetate over the

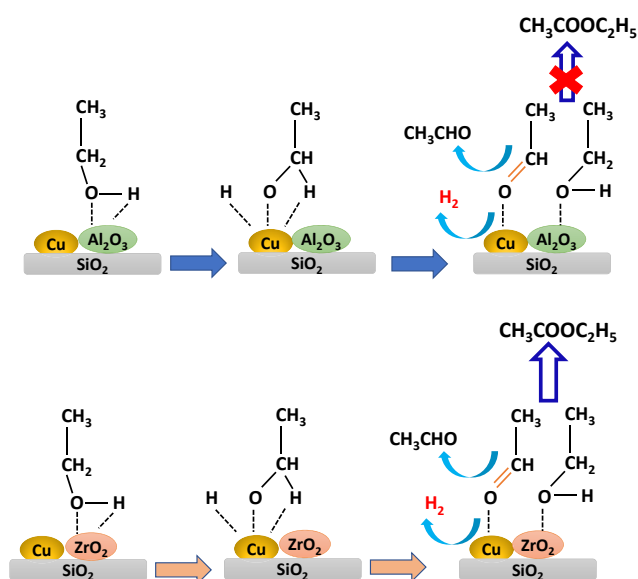


Fig. 3 – Reaction pathways of ethanol dehydrogenation and dehydrogenative coupling over $\text{Cu}/\text{SiO}_2\text{--Al}_2\text{O}_3$ and $\text{Cu}/\text{SiO}_2\text{--ZrO}_2$. Reproduced from Ref. [38].

Cu-based catalyst samples with a more heterogenous distribution of copper species (Cu^0/Cu^+) [60]. Cassinelli et al. reported that the addition of Cu^+ enhanced ethanol dehydrogenation than that of using only Cu^0 [77]. Recently, Ob-eye et al. [62] reported that Cu^+ is an active site for acetaldehyde production than that of Cu^0 , while Cu^0 is active for ethyl acetate production [60]. Suggesting that the yield of hydrogen can increase with the increase of coupling reaction of ethanol with acetaldehyde to form ethyl acetate. Therefore, the interaction of copper-support is important to stabilize Cu^+ and Cu^0 to tailor the formation of hydrogen, acetaldehyde, and ethyl acetate.

Effect of metal loading

The percentage of Cu on support strongly impacts ethanol conversion and product selectivity [35,60,76,78], the Cu loading can be separated into low loading (<10%) and high loading (>10%). The effect of Cu loading is different depending on the support and reaction temperature condition. The high reaction temperature favors high ethanol conversion (Fig. 4a and b). In the case of $\text{Zn}/\text{ZnAl}_2\text{O}_4$ support [78], ethanol conversion increases with an increase of copper loading from 0% to 20% (Fig. 4a) due to the increase of density of active copper sites on the surface of the support. Similar support (ZnAl_2O_4) [35] gives the highest ethanol conversion with 30% copper metal loading and decrease after that because the interaction of Cu species with sub-stoichiometric spinel decreases with Cu loading increases [35], resulting in low activity at high copper loading. For ZrO_2 support, 10% of copper metal loading shows the highest ethanol conversion. While SiO_2 shows high ethanol conversion at Cu loading from low to high (50%) copper loading [76]. That is due to the dispersion of Cu on the

surface of supports as well as the interaction of Cu metal with support.

In the case of SiO_2 support, the metal loading is higher than 2.5 wt% enhances the ethanol conversion (Fig. 4c), however, the productivity decrease with an increase of Cu loading. Indicating that very high Cu loading is not necessary for this reaction [76]. This trend is a strong relationship with the dispersion of Cu on the surface of the catalyst, indeed, Cu dispersion decreases from 75.3% (for 0.5% Cu) to 33.6% (for 20% Cu) on SiO_2 -AE support. Additionally, small particles are more reactive because they contain a higher density of coordinatively unsaturated sites (cus) such as corners and kinks [76,79]. Besides that, the low Cu loading can show high productivity, it can deactivate easily due to the sintering of Cu causing a loss of the catalyst activity. Therefore, Cu loading is required enough to avoid quick sintering, such as > 2.5% in the case of Cu/SiO_2 [76].

Looking at the product selectivity (Fig. 4d), the selectivity of ethyl acetate increases with an increase of acetaldehyde, confirming that acetaldehyde is an intermediate of ethyl acetate. In the case of ZrO_2 support, ethyl acetate gives high selectivity (70–75%) at 10 wt% Cu loading, while acetaldehyde has 15–20% selectivity [60]. For SiO_2 support, acetaldehyde is the main product, and ethyl acetate is a minor product [38,76]. Suggesting that not only the effect of Cu loading but the ethanol conversion and product selectivity are also influenced by the support. In fact, ZrO_2 support can interact with Cu metal to give a high density of basic sites of O^{2-} ion and a more heterogeneous distribution of Cu species (Cu^0/Cu^+) resulting in high ethyl acetate selectivity [60]. Additionally, Finger et al. [38] found that the heterogeneous distribution of Cu species (Cu^0/Cu^+) increases with the addition of ZrO_2 into $\text{Cu}/\text{mesoporous silica MCM-41}$ catalyst. Suggesting that the interaction

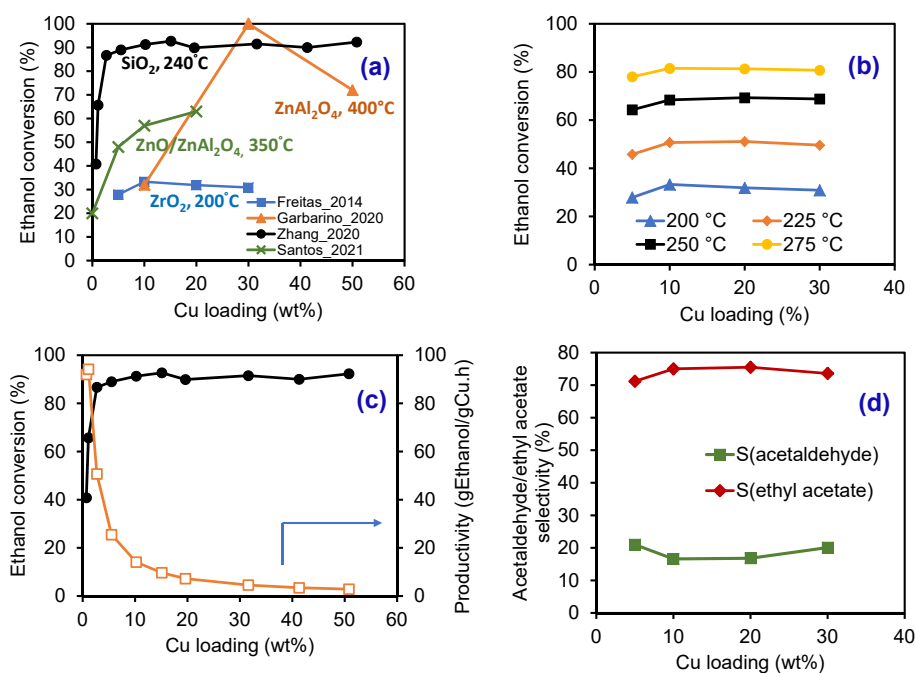


Fig. 4 – Ethanol conversion (a,b); ethanol conversion and productivity (c); and selectivity to acetaldehyde and ethyl acetate as a function of Cu loading (d). Reproduced from Refs. [35,60,76].

of Cu with ZrO_2 and SiO_2 are different, resulting in the difference of product selectivity.

Effect of particle size and dispersion Cu particles

The particle size and dispersion of Cu particles are the important factors affecting the catalytic conversion and product selectivity of ethanol dehydrogenation. Fig. 5a showed that the small particle size gives high ethanol conversion using Cu/silicon carbide support [61]. The same manner is also observed over different catalysts, Cu/ SiO_2 , Cu/PPC and Cu/silicon carbide, with high ethanol conversion >80% at Cu particle size <10 nm (Fig. 5c) [61,76,80,81]. That can be explained that small Cu particles possess a higher density of coordinatively unsaturated sites, e.g. corners and kinks, which are more reactive than steps or terrace sites [76,79]. However, at low Cu particle size (<10 nm), the ethanol conversion is also low (<40%) over some catalysts due to the effect of support to Cu-based catalysts. Indeed, as the same small Cu particle size (<10 nm), the mesoporous support (61% ethanol conversion over Cu/ MgAlO_x -P) shows high catalytic activity than that of non-porous support (28% ethanol conversion over Cu/ MgAlO_x) [82]. Moreover, SiO_2 support exhibits higher ethanol conversion (85.4% ethanol conversion) than that of carbon support (17.4% ethanol conversion) [80]. Looking at the dispersion of Cu particles, the dispersion of Cu particles in the range of 20%–60% gives high ethanol conversion (Fig. 5d). Generally, the high Cu dispersion will provide a high amount of active sites resulting in high ethanol conversion. But with low Cu loading leads to low total Cu active sites and low

ethanol conversion. In Fig. 5a, the high Cu dispersion showed high ethanol conversion (81.4%) in the case of Cu/ SiO_2 /SiC catalyst. This trend is relative to the high active sites resulting from the high dispersion of Cu species on the surface of SiO_2 -rich surface [61]. With low Cu dispersion (3.5% for Cu/C/SiC catalyst; and 5.7% for Cu/SiC), the lower dispersion of Cu species (3.5%) shows high ethanol conversion than that of middle Cu dispersion (5.7%) due to the C-rich surface accelerates the enrichment of ethanol concentration resulting in high ethanol conversion [61].

The different trends in particle size and dispersion can be presented by TOF over Cu/0.675Al–Zr and Cu/Zr catalyst (Fig. 5b). TOF of ethanol over Cu/Zr catalyst increases with an increase of particle size and a decrease of Cu dispersion, whereas that trend is reverse using Cu/0.675Al–Zr catalyst [72]. That indicates that the interface dominates the catalyst activity rather than the Cu active sites [72].

For the product selectivity, small particle size usually favors acetaldehyde formation, while larger copper particles can favor ethyl acetate production [82]. Therefore, the particle size of Cu particles can be adjusted to select the desired products.

In 2020, Zhang et al. [76] showed a promising Cu/ SiO_2 catalyst with >90% ethanol conversion and >95% acetaldehyde selectivity. Cu/ SiO_2 with 10 wt% Cu showed well dispersion of Cu particles (38.7%) and small particle size (2.7 nm) using the ammonium evaporation method. The HRTEM image showed that no CuO particles were observed even at 5 nm resolution at 600 °C calcination (Fig. 6). Suggesting the ammonia evaporation method is an excellent method to prepare a high dispersion of Cu on the support.

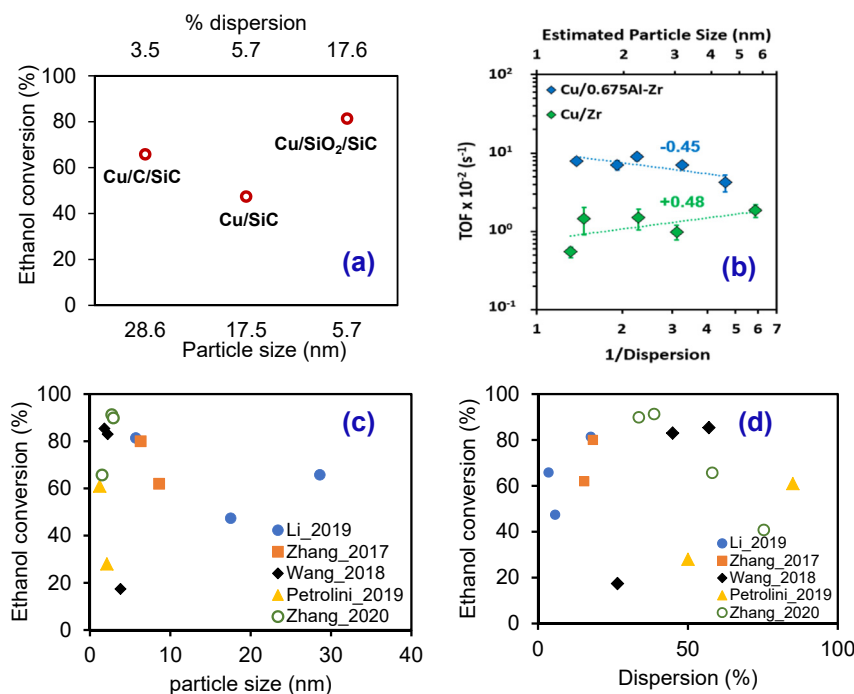


Fig. 5 – Ethanol conversion as a function of particle size and dispersion (a); Turnover frequency (TOF) as a function of particle size and 1/dispersion (b); ethanol conversion as a function of particle size (c) and dispersion (d). Reproduce from Refs. [61,72,76,80–82].

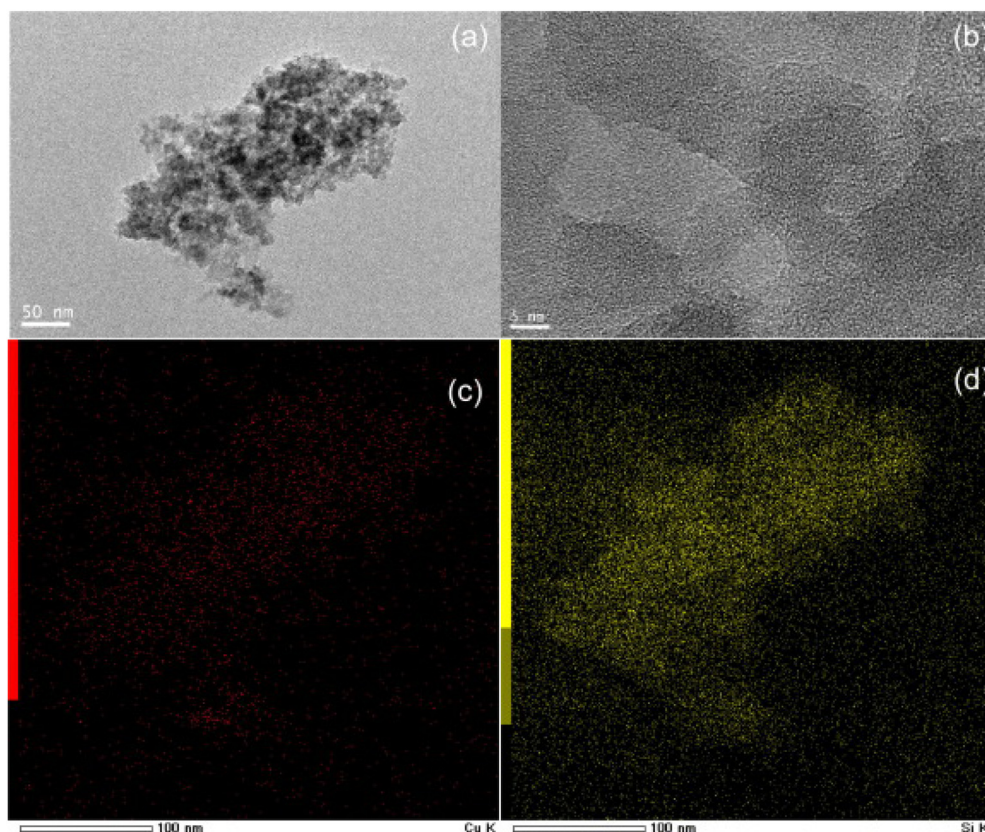


Fig. 6 – Transmission electron micrographs (a, b) and elemental maps of Cu and Si (c, d) of 10 wt% Cu/SiO₂-AE. Reproduce from Ref. [76].

Support

As discussed in the effect of Cu particles, the support can provide a positive or negative effect on the catalyst. In fact, the interface of the Cu-support is very important in tailoring the products because different support can direct different reaction pathways and the catalyst activity.

The support with high acidity favors ethanol dehydration into ethylene, while ethanol dehydrogenation takes place on weak acid or basis sites [76,83,84]. Phung et al. [85] reported that catalysts possessing both acid and base sites (MgO–Al₂O₃) showed high acetaldehyde selectivity than that of ZrO₂, SiO₂, and TiO₂. While the strong Lewis acid γ -Al₂O₃ did not observe acetaldehyde and mainly ethanol dehydration products. In addition, Zhang et al. [76] reported that Cu/SiO₂ showed the best ethanol conversion ($X > 90\%$) and acetaldehyde selectivity ($S > 98\%$) at 240 °C in comparison to Cu/ZrO₂ ($X = 52\%$, $S = 86\%$), Cu/La₂O₃ ($X = 43\%$, $S = 98.5\%$), Cu/MgO ($X = 60\%$, $S = 98.5\%$), and Cu/MSU-9 ($X = 78\%$, $S = 52\%$; MSU = mesoporous silica-alumina Si/Al = 9).

The support can enhance the ethanol conversion and ethanol dehydrogenation reaction to form hydrogen, acetaldehyde, and further reaction to ethyl acetate. Good support improves the dispersion of Cu on the surface of the catalyst leading to an increase in acetaldehyde selectivity and stability [86]. Indeed, hybrid C@SiO₂ hollow submicron spheres showed a high dispersion of Cu species due to the highly dispersed

silica species among carbons [86]. The Cu/C@SiO₂ performed 36.1% ethanol conversion and ~99% acetaldehyde selectivity at 260 °C. Additionally, the high dispersion of Cu on the surface of the hybrid carbon-silica prevents the sintering and agglomeration of copper particles during the reaction process resulting in the improvement of the catalyst stability.

Not only the high dispersion of Cu species on the support but also the interaction of Cu species with support and the interaction of intermediate and product with support are important [61]. For example, ethyl acetate is formed a high yield on the surface of Cu/ZrO₂, but not on the surface of Cu/SiO₂ + ZrO₂ [87], confirming the role of the interface Cu-support on the dehydrogenative coupling reaction. Recently, Finger et al. [38] performed the effect of metal oxides on Cu-based catalysts, they found that the rate of ethyl acetate increases with the addition of metal oxides with the trend: no-metal oxides adding $< \text{Cr}_2\text{O}_3 < \text{Al}_2\text{O}_3 < \text{ZrO}_2 = \text{ZnO}$. The difference in ethyl acetate rate is due to the effect of the interface Cu-support. In fact, the metal oxides support can tailor the agglomeration of Cu nanoparticles showing different Cu dispersion [38]. Also, the addition of metal oxides can also introduce different Cu⁺ and Cu⁰ on the pristine 10Cu/MCM-41, e.g., the ratios of Cu⁺ δ -CO/(Cu⁺ δ -CO + Cu⁰ δ -CO) are 0.55 (for 5ZrO₂–10Cu/MCM-41) and 0.78 (for 5ZnO–10Cu/MCM-41), in compare with 0.32 of 10Cu/MCM-41 sample. Whereas the addition of Al₂O₃ showed the strong interaction of Cu–O–Al meaning that it possesses low electron density sites. In fact, different metal oxides support shows different TOF of ethyl

acetate (Fig. 7a), where ZnO and ZrO₂ show the highest TOF of ethyl acetate [38]. Confirming that the dehydrogenative coupling reaction depends on the activation of ethanol by the interface of copper with metal oxides. For the metal and metal oxide loading effects, the ethanol conversion increased with an increase of ZrO₂ loading in comparison among 2ZrO₂-10Cu/MCM-41, 2ZrO₂-15Cu/MCM-41 and 5ZrO₂-10Cu/MCM-41 [38]. In which, TOF of ethyl acetate increases with an increase of ZrO₂/Cu ratio (Fig. 7b), indicating that the rate of dehydrogenative coupling is controlled by the interface of copper-metal oxide [38].

Petrolini et al. reported that ethanol dehydrogenative reactions over copper supported on porous Al–Mg mixed oxides [82], they found that mesoporous support can enhance the ethanol conversion and acetaldehyde and ethyl acetate selectivity. In fact, the mesoporous support enhances the dispersion of copper (85%) in comparison to the non-porous support (50%). Additionally, XANES spectra also showed that the Cu²⁺ reduced to Cu⁰ with a small amount of intermediate Cu⁺ over Cu/MgAlO_x (non-porous support) and 25% of Cu⁺ over Cu/MgAlO_x-P (porous support), that due to the monolayer dispersion of Cu on the surface of porous support [82]. Suggesting that mesoporous support is essential for the dispersion of copper, which can decide the acetaldehyde and ethyl acetate production. Looking at the product selectivity, to increase the selectivity to ethyl acetate from <2% at WHSV = 31.1 h⁻¹, the contact time has to increase up to 6.6 times (WHSV = 4.7 h⁻¹) [82]. At this contact time (4.7 h⁻¹), the non-porous Cu/MgAlO_x catalyst showed a higher selectivity to ethyl acetate (35%) than that of porous Cu/MgAlO_x-P catalyst at the first cycle in time-on-stream of 6 h. After regenerating the catalysts, ethyl acetate selectivity increases in both catalysts and reaches around 40% selectivity due to the increase of Cu particles size (2.1 and 2.5 nm). While the selectivity to acetaldehyde decreases in both catalysts after regeneration, but the selectivity to acetaldehyde over the porous catalyst is always higher than that of a non-porous catalyst. For ethanol conversion, the porous catalyst showed higher conversion (~90%) than that of non-porous catalyst (<70% and decrease following time increases). That reveals that the selectivity to ethyl acetate and acetaldehyde rely on the Cu particle size and the interface of copper-support [82]. In other words, the increase of copper-support interface leads to the improvement

of ethyl acetate selectivity, or larger copper particles can favor ethyl acetate production.

Promoter

Cu showed as a good catalyst for ethanol dehydrogenation, however, it still has a drawback due to the sintering at high temperatures. To avoid the sintering of Cu, co-adding metal or promoter can be used such as Ni [59,75,88], Ag [89,90], and Pt [91]. For Ag, it is controversial due to Ag can be a promoter [90] or inhibitor by blockage Cu active surface by Ag particle [89]. Therefore, it needs to study more about the effect of Ag in ethanol dehydrogenation. Pt is also a candidate for promoting the activity of Cu. Wang et al. [91] reported that Pt–Cu single atom alloys enhance sixfold the activity of Cu (111) for ethanol dehydrogenation. Pt can support the O–H bond scissors followed by ethoxy spillover to the Cu (111) sites, then the C–H bond scissors to form acetaldehyde [91].

Interestingly, copper and nickel are well-known to form an alloy, and in period 4 of the periodic table. They are close together (Z = 28 and Z = 29) and have a similar atomic radius (124 p.m. vs 128 p.m.) and FCC lattice structure. Additionally, Ni can protect Cu particles due to the higher melting point (1455 °C) in comparison to that of copper (1084 °C), making their alloy less mobile [59]. De Waele et al. [59] reported that 1 wt% Ni addition into Cu (10 wt% and 20 wt%; samples 1Ni–10Cu and 1Ni–20Cu) showed similar and higher ethanol conversion in comparison to sole Cu. 1Ni–10Cu sample has ~ double turn-over frequency with respect to sole Cu (10 wt%). Suggesting that the addition of Ni (1 wt%) enhances the catalytic activity of Cu-based catalysts. Moreover, 1Ni–10Cu is also stable after a 40 h operation with a reaction condition at 260 °C, 0.5 MPa, a space-time of 3.6 kg_{cat}/s/mol and a N₂/ethanol-ratio of 20. Moreover, the formation of acetaldehyde occurred at a lower temperature (150 °C) over Ni–Cu alloy, while sole Cu/SiO₂ is unreactive up to 250 °C [75]. The activation of Ni–Cu alloys occurs at low temperature because the presence of Ni lowers the activation energy barrier of the cleavage of the C–H bond, which is found as a rate-determining step in the ethanol dehydrogenation [75]. Shan et al. [75] also found that Ni–Cu alloys are active than Pt–Cu and Pd–Cu alloys in ethanol dehydrogenation due to Ni

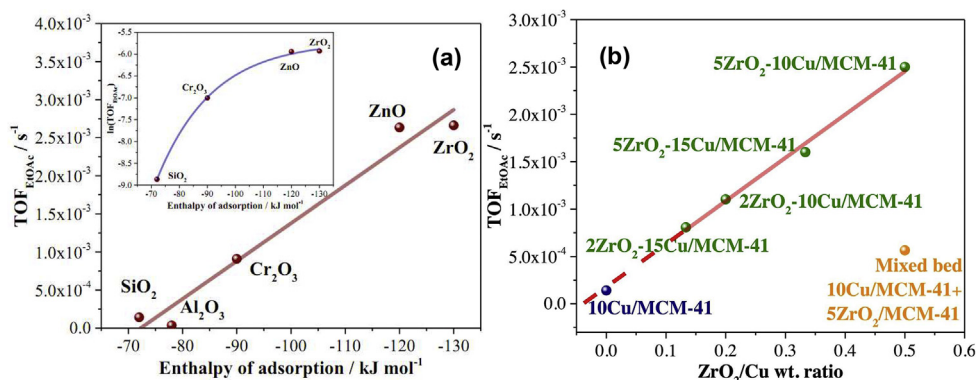


Fig. 7 – Turnover frequency (TOF) of ethyl acetate as a function of ethanol enthalpy of adsorption (a); as a function of ZrO₂/Cu wt ratio (b). Reproduce from Ref. [38].

causes the lower apparent activation energy ($E_a = 45\text{--}49$ kJ/mol) with respect to that of Pd (69 kJ/mol) and Pt (63 kJ/mol).

The addition of Ni also improves the stability of catalysts, 1Ni–10Cu showed the lowest carbon deposit (5.23 mol/g) in comparison to other Ni–Cu alloy and sole Cu or Ni [59]. Indeed, sole Ni showed a very high carbon deposit (25.48 mol/g), the co-presence of metals (Cu and Ni) will reduce the coke deposition. However, the low amount of Cu (5 wt%) gave a low effect than that of a high amount of Cu (10–20 wt%). That is possibly due to the less effect of Ni with a high amount of Cu [59]. Additionally, ethanol is adsorbed to form acetaldehyde on the major Cu sites via η^1 -acetaldehyde transition state [70] instead of η^2 -acetaldehyde over Ni metal, which further converted into CO and CH₄, then carbon [92,93]. Also, adding a small amount of Ni does not change the coordination number and bond length of Cu–Cu, but interestingly the local charge transfer between dispersed Ni and Cu atoms help to improve the stability of catalysts due to against sintering at high temperature [75,88].

Reaction conditions

Besides the effect of copper-based catalysts, acetaldehyde and ethyl acetate productions are influenced by the reaction condition [38]. The reaction rate for acetaldehyde decreases with an increase of contact time (W/F) over mostly catalysts (Fig. 8a), this is due to the further reaction of acetaldehyde and ethanol into ethyl acetate through the dehydrogenative coupling reaction [38]. Whereas the reaction rate for ethyl acetate gets maximum at different W/F, e.g., $2.6 \times 10^{-3} \text{ s}^{-1}$ for 5ZrO₂–10Cu/MCM-41 and 5ZnO–10Cu/MCM-41 (Fig. 8b). The reaction rate for ethyl acetate decreases with an increase of contact time (except for mixed oxides) (Fig. 8b) possibly due to the deformation of ethyl acetate into other products.

For reaction temperature, the higher temperature enhances ethanol conversion and TOF over different catalysts (Fig. 9). However, the high temperature usually causes deactivation catalyst and formation of by-products Cu/ZrO₂ catalyst, we observed the high product selectivity of ethyl acetate and low selectivity to acetaldehyde at 250 °C (Fig. 9b).

Where we can observe high hydrogen selectivity, which can calculate by the sum of selectivity of detected acetaldehyde and the twice of detected ethyl acetate (twice detected ethyl acetate = consumed acetaldehyde involving in coupling reaction + produced ethyl acetate).

Moreover, bioethanol from fermentation process contains a high amount of water. Separating water by distillation and membrane filtration into pure ethanol for the conversion will increase the cost of the process. If we can directly transform the feed of aqueous ethanol into hydrogen, acetaldehyde, and ethyl acetate without any separation will reduce the cost of the process and improve the use of different raw materials. De Waele et al. [59] reported that the presence of 30 wt% H₂O/ethanol not only was not harmful to the ethanol conversion but so enhance the ethanol conversion in comparison to pure ethanol feed over Ni–Cu alloys. That can be explained that the presence of water can help to reduce the deactivation of catalyst causing by carbon deposit by the formation of H₂ and O* on the surface of Cu metal, which can remove the carbon deposit on the surface of catalysts [59,94].

Looking at the rate of ethanol consumption (Table 1), which is calculated by [flowrate of ethanol feedstock (kg/h) * Ethanol conversion (%)]/[catalyst (kg) * Cu loading (%)], Cu/MgAlO_x [82], Cu/SiO₂/SiC and Cu/C/SiC [61], and 5Cu/ZrO₂ [60] show highest ethanol consumption rate. Suggesting that those catalysts are among the best catalysts for ethanol dehydrogenation. However, both Cu/SiO₂/SiC and Cu/C/SiC catalysts need to improve the Cu dispersion by possibly applying proper preparation techniques to improve the catalytic activity per active site. While Cu/MgAlO_x needs to improve the stability to reduce the decrement of Cu dispersion and to maintain ethanol conversion. Additionally, based on the data of ethanol consumption rate allows us to have an overview of the activity of catalysts and adjust the reaction condition to get a high ethanol consumption rate. In fact, several catalysts gave high ethanol conversion and product selectivity but exhibited a low ethanol consumption rate, such as HT-Cu40 (entry 8) and Cu-MFI-AE (entry 24). That is due to the application of a low ethanol flowrate. The rate of ethanol consumption can be examined by modification of reaction conditions to obtain the desired rate.

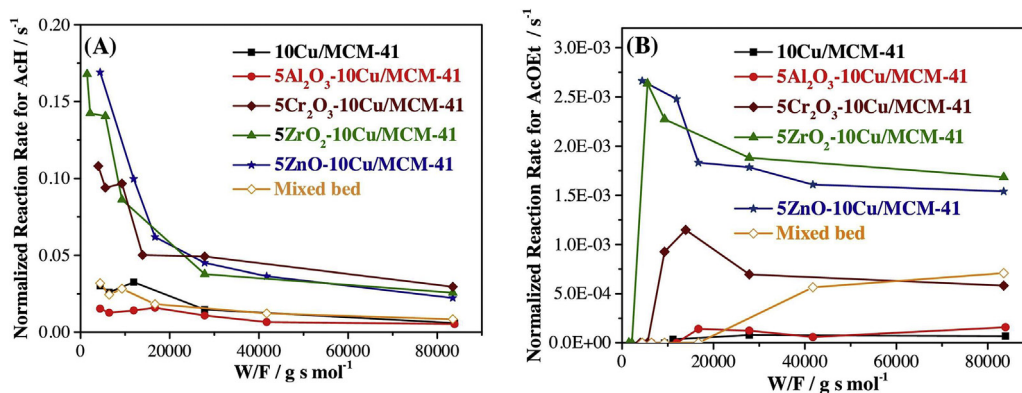


Fig. 8 – Normalized reaction rate for acetaldehyde (A) and ethyl acetate (B) formation as a function of the contact time (W/F). Reproduce from Ref. [38].

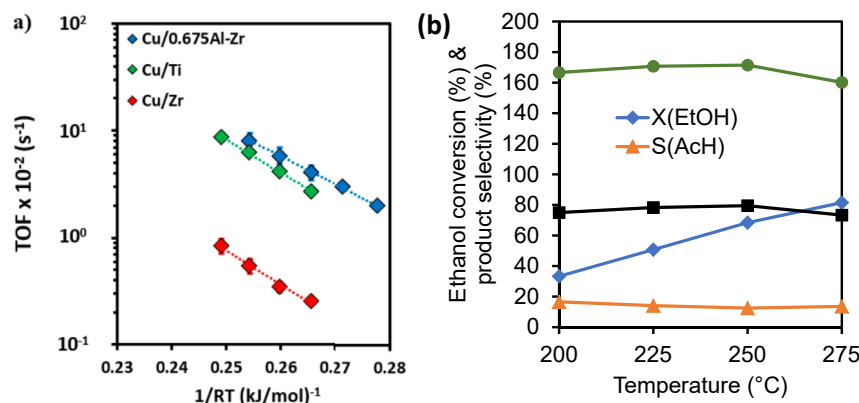


Fig. 9 – Ethanol turnover frequency (TOF) at 71.5 mbar ethanol as a function of $1/\text{temperature}$ (a); and ethanol conversion and product selectivity as a function of temperature (b). Reproduce from Refs. [60,72].

Stability

The stability of a catalyst is one of the most important factors of a successful catalyst. The stability of catalysts can decide how application of catalysts in the industry rather than its high activity due to economic reasons. For Cu/support catalyst, the deactivation is due to the sintering of Cu particles and coke formation on the surface of catalysts [60,76]. Rosset and Perez-Lopez found that the catalyst with lower Cu content is deactivated mainly by carbon deposition, whereas the catalyst with higher Cu content is deactivated by sintering [73,95]. To avoid the sintering of Cu particles, it needs to increase the dispersion of Cu on the surface of support as well as enhance the interaction of Cu with supports.

Zhang et al. [76] reported the ammonia evaporation method showed high dispersion of Cu particles and high thermal stability due to the formation of the copper phyllosilicate phase. The formation of copper phyllosilicate ($\text{Cu}_2\text{Si}_2\text{O}_5(\text{OH})_2$) was confirmed by XRD patterns of Cu/ SiO_2 and FT-IR spectra (Fig. 10). The copper phyllosilicate phase was also confirmed with the peak at 670 cm^{-1} and 1032 cm^{-1} assigned to the distinctive δ_{OH} and ν_{SiO} , respectively (Fig. 10a $\rightarrow$ c). Those bands increased with an increase of Cu loading and disappeared after calcination at $600\text{ }^{\circ}\text{C}$ due to the decomposition into CuO and SiO_2 [76]. While XRD patterns confirmed the presence of the copper phyllosilicate phase (Fig. 10d). Additionally, the deactivation of Cu-based catalysts also depends on the morphology of support. Cu/ MgAlO_x catalysts are deactivated due to the sintering of Cu particles and are less prominent on the porous catalysts (Cu/ $\text{MgAlO}_x\text{-P}$) than that of non-porous catalysts (Cu/ MgAlO_x) because of the physical barrier that Cu particles are well-dispersed on the macro and mesoporous support [82]. Regarding the preparation method, mixed metal oxides synthesized from salt precursor showed homogeneous dispersion of Cu on the support enhancing the stability of catalyst in comparison to the catalysts prepared using conventional impregnation methods [78]. While the crystallized MFI support showed the strong interfacial interactions between Cu species and support via the Cu–O–Si bonds preventing the reduction of Cu^+ and inhibited the Cu^0 aggregation resulting in the high stability of Cu-MFI-AE catalyst [74].

Discussion and outlook

As mentioned previously in the introduction and reaction mechanism section, hydrogen can be produced through ethanol dehydrogenation and dehydrogenative coupling of ethanol with acetaldehyde. Therefore, the yield of hydrogen can be controlled by what reaction occurs. In fact, the yield of hydrogen increases with the ethyl acetate formation, because the first hydrogen is produced along with acetaldehyde through ethanol dehydrogenation, and the second hydrogen is produced along with ethyl acetate formation through the dehydrogenative coupling reaction. It means that the yield of total hydrogen is equal to the yield of acetaldehyde detected plus twice the yield of ethyl acetate detected.

There are many factors that impact the formation of hydrogen, acetaldehyde, and ethyl acetate including the Cu particles, support, reaction condition, promoter, and feedstock. The summary is illustrated in Fig. 11. The particle, dispersion, and loading of Cu strongly influence ethanol dehydrogenation and product selectivity. The range of Cu loading should be around 10 wt% for enough active sites and avoiding sintering deactivation, it should not be lower and higher due to the deactivation at lower Cu loading and agglomeration with high Cu loading. For the particle size, the low particle size ($<10\text{ nm}$) favors acetaldehyde production, but the large particle size favors ethyl acetate. The dispersion of Cu should be in the range of 20–60% to achieve a high ethanol conversion and the stability of the catalyst as well.

For the support, the strong interface of Cu-support is very important to avoid Cu sintering and ethyl acetate formation. Also, support can play an important role in product selectivity. Table 2 summarized the advantages and limitations of several potential supports in Cu-based catalysts for ethanol dehydrogenation. The high acetaldehyde can be obtained with low acidity of support, such as SiO_2 [76], while ethyl acetate needs the presence of other supports such as ZrO_2 or ZnO [38,60]. That allows controlling the yield of acetaldehyde and ethyl acetate by using a mixture of support, e.g. ZrO_2 and SiO_2 with a defined ratio. Besides that, the ratio of acid sites to the basicity of catalyst also affects catalytic ethanol conversion. The high basicity does not favor ethanol dehydrogenation reaction in

Table 1 – The summary of ethanol dehydrogenation and dehydrogenative coupling with high ethanol conversion in over Cu-based catalysts.

Entry	Catalyst	Support	Catalyst mass (g)	Cu loading (wt%)	Cu dispersion (%)	Particle size (nm)	WHSV (h^{-1})	Time (min.)	Temp. ($^{\circ}\text{C}$)	Ethanol conv. (%)	Rate of ethanol consumption (kg EtOH/(kg Cu *h))	S(AcH) (%)	S (EA) (%)	S(H_2) (%)	Ref.
1	Cu/PPC	PPC	0.1	10	18.3	6.3	2.4	100	180	80	19.2	~95	—	~95	[81]
2	Cu/CMC	CMC	0.1	10	15.4	8.6	2.4	100	180	60	14.4	~95	—	~95	
3	Cu20CA-C	Ca/Al	0.2	20	—	—	1.6	—	275	92	7.3	92	—	92	[95]
4	Cu40CA-C	Ca/Al	0.2	40	—	—	1.6	—	275	94	3.7	92	—	92	
5	Cu20CA-R	Ca/Al	0.2	20	—	—	1.6	—	275	88	6.9	94	—	94	
6	Cu40CA-R	Ca/Al	0.2	40	—	—	1.6	—	275	96	3.8	93	—	93	
7	HT-Cu20	Mg/Al	0.2	20	—	—	1.6	—	275	94.2	7.4	97.2	0.4	98.0	[73]
8	HT-Cu40	Mg/Al	0.2	40	—	—	1.6	—	275	97.8	3.9	92.3	2.8	97.9	
9	C–Cu20	Mg/Al	0.2	20	—	—	1.6	—	275	96.1	7.6	92.3	2.6	97.5	
10	C–Cu40	Mg/Al	0.2	40	—	—	1.6	—	275	95.6	3.8	87.1	5.6	98.3	
11	R–Cu20	Mg/Al	0.2	20	—	—	1.6	—	275	89.7	7.1	92.1	2.9	97.9	
12	R–Cu40	Mg/Al	0.2	40	—	—	1.6	—	275	95.8	3.8	87.8	6.9	101.6	
13	Cu/MgAlO _x -P	MgAlO _x	0.15	10	85	1.2	31.1	—	300	61	189.7	98	—	98	[82]
14	Cu/MgAlO _x	MgAlO _x	0.15	10	50	2.1	31.1	—	300	28	87.1	94	—	94	
15	Cu/MgAlO _x -P	MgAlO _x	0.24	10	85	1.2	4.7	360	300	90	42.3	~75	~20	115	
16	Cu/MgAlO _x	MgAlO _x	0.24	10	50	2.1	4.7	360	300	70	32.9	~60	~35	130	
17	Cu30ZA	ZnAl ₂ O ₄	—	30	—	—	—	60	400	~100	—	~90	~5	100	[35]
18	Cu/C/SiO ₂	C/SiO ₂	0.1	8.8	44.9	2.2	2.4	600	260	83	22.6	95.1	—	95.1	[80]
19	Cu/SiO ₂	SiO ₂	0.1	8.4	57.0	1.8	2.4	600	260	85.4	24.4	79.4	—	79.4	
20	Cu/SiC	SiC	0.1	5	5.7	17.5	2.4	—	280	47.4	22.8	97.7	—	97.7	[61]
21	Cu/SiO ₂ /SiC	SiO ₂ /SiC	0.1	5	17.6	5.7	2.4	—	280	81.4	39.1	94.2	—	94.2	
22	Cu/C/SiC	C/SiC	0.1	5	3.5	28.6	2.4	—	280	65.8	31.6	99.0	—	99.0	
23	Cu/SiO ₂ -AE	SiO ₂	0.1	10	38.7	2.7	1.6	—	300	~100	15.8	~98	—	98	[76]
24	Cu-MFI-AE	MFI	—	19.8	23.6	—	0.5	—	250	97	2.6	93	—	93	[74]
25	5Cu/ZrO ₂	ZrO ₂	0.5	5	34	3.1	3.2	—	275	78.0	49.3	16.1	68.3	152.7	[60]
26	10Cu/ZrO ₂	ZrO ₂	0.5	10	23	4.4	3.2	—	275	81.5	25.7	13.6	73.3	160.2	
27	20Cu/ZrO ₂	ZrO ₂	0.5	20	14	7.6	3.2	—	275	81.3	12.8	13.2	76.2	165.6	
28	30Cu/ZrO ₂	ZrO ₂	0.5	30	8	12.6	3.2	—	275	80.7	8.5	15.0	73.6	162.2	

PPC = polyacrylonitrile-based porous carbon; CMC = Commercial microporous carbon; S(AcH) = selectivity to acetaldehyde; S(EA) = selectivity to ethyl acetate; S(H_2) = selectivity to hydro; WHSV = weight hour space velocity.

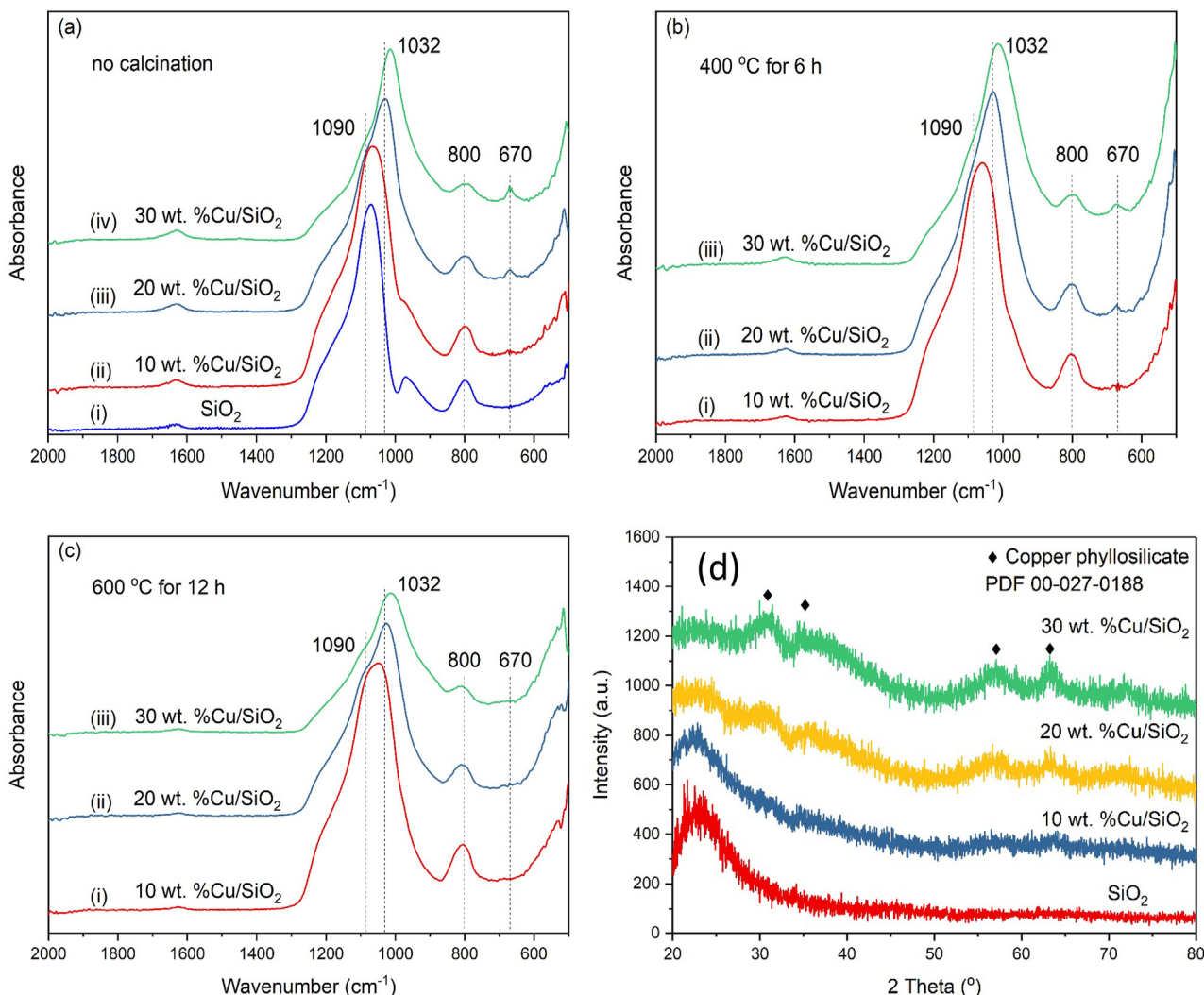


Fig. 10 – ATR-FTIR spectra of Cu/SiO₂-AE with different Cu loading before calcination (a) and calcined at 400 °C (b) and 600 °C (c); and X-ray diffractograms of Cu/SiO₂-AE with different Cu loading before calcination (d). Reproduce from Ref. [76].

the case of Co/ACC catalyst, while the highest ethanol conversion and acetaldehyde selectivity on Cu/ACC, the highest acid catalyst and low basicity catalyst (Fig. 2) [62]. In addition, the catalysts with porous structure will enhance the dispersion of Cu resulting in the improvement of stability and Cu dispersion. In the case of the promoter, 1 wt% Ni is a potential candidate for the improvement of ethanol conversion and the stability of the Cu-based catalyst due to the decrease of activation energy resulting in the decrease in reaction temperature and operation cost.

Moreover, reaction conditions, such as reaction temperature, contact time as well as the pressure and purity of ethanol feedstock, impact the conversion and selectivity. The high reaction temperature often enhances the ethanol conversion, but also produces by-product or coke formation. High contact time can enhance the formation of ethyl acetate, which needs time to form by coupling ethanol with acetaldehyde produced. While the pressure of ethanol can impact the reaction mechanism. Interestingly, the presence of water in the feedstock can improve the yield of product opening an important

point to use aqueous ethanol solution as feedstock instead of pure ethanol resulting in lowering production cost.

Looking at the several potential Cu-based catalysts for ethanol dehydrogenation and dehydrogenative coupling (Table 1), support containing both acid and base properties including Mg/Al [73], Ca/Al [95] and ZnAl₂O₄ [35] showed higher ethanol conversion (>95%) and acetaldehyde selectivity (entries 6, 8–10, 12, 15, and 174). Taking into account absolute values of conversion of ethanol, for the low acidity support, the new ammonia evaporation method is a promising method to produce a strong interaction of Cu with SiO₂ support resulting in high ethanol conversion (~100%) and acetaldehyde selectivity (~98%) (entry 23) [76]. Cu/SiO₂-AE catalyst is a potential candidate to develop for the further studies. Additionally, the ammonia evaporation method with an MFI structure of silica support (Cu/MFI-AE catalyst) shows high acetaldehyde selectivity (93%) and high ethanol conversion (97%) at 250 °C (entry 24) [74]. Especially, Cu/MFI-AE exhibits high stability with high ethanol conversion and acetaldehyde selectivity for 500 h time-on-stream [74].

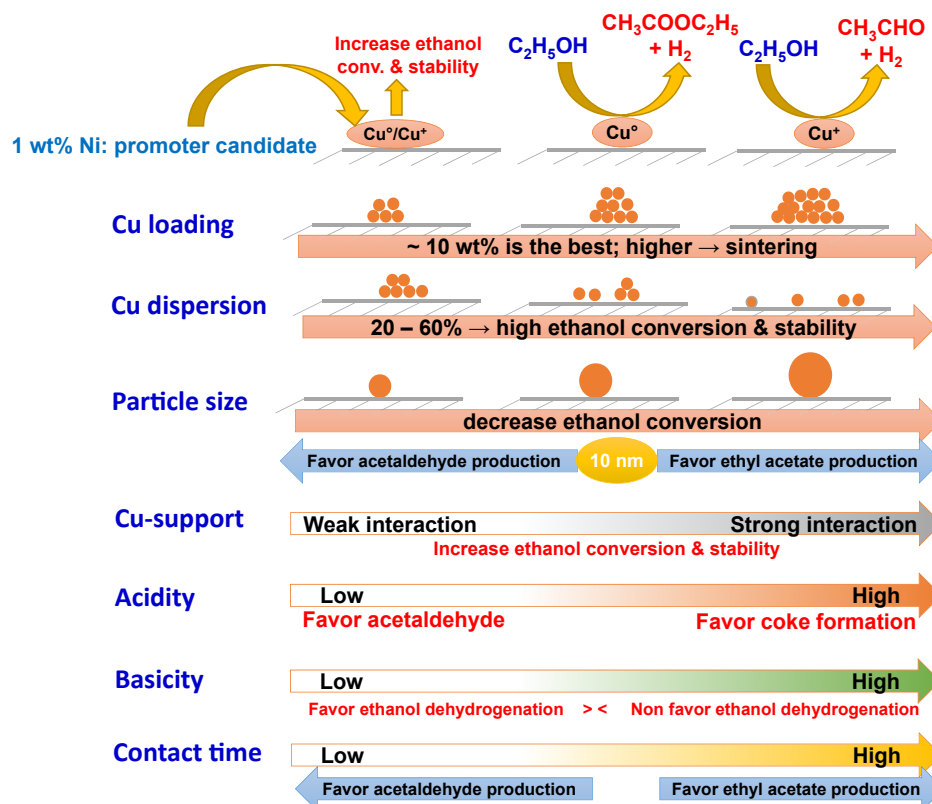


Fig. 11 – The scheme of the summary of all parameters affect ethanol dehydrogenation.

Table 2 – The advantages and limitations of several potential supports in Cu-based catalysts for ethanol dehydrogenation.

Entry	Support	Advantages	Limitations
1	SiO ₂	- High ethanol conversion - High selectivity to acetaldehyde - Strong interaction with copper	- Need specific preparation method to prevent the deactivation. - Need adding other material, such as C, to prevent the sintering and agglomeration of copper.
2	Mesoporous silica	- High ethanol conversion - High selectivity to acetaldehyde - Strong interaction with copper	Low selectivity to ethyl acetate
3	Silicon carbide	- High interaction with copper - High ethanol consumption rate	- Low Cu dispersion and activity when using solely. - Need co-support, i.e. C or SiO₂, to improve activity. - Complex preparation.
4	Zeolite-structure silica	- High ethanol conversion - High selectivity to acetaldehyde - Strong interaction with copper	Low selectivity to ethyl acetate
5	ZrO ₂	- Strong interaction with copper - Favor to form ethyl acetate - High ethanol consumption rate	Low ethanol conversion that SiO₂
6	Al–Mg mixed oxides	- Producing both acetaldehyde and ethyl acetate - High ethanol consumption rate	- Need mesoporous structure - Need high temperature reaction. - Huge reduction of Cu dispersion.

Combining with the effect of ZrO₂ support reported by Finger et al. [38] and Freitas et al. [60], the selectivity of acetaldehyde and ethyl acetate can be controlled by the amount of ZrO₂ addition into Cu/SiO₂-AE catalyst. Also, Ni (~1 wt%) can add to Cu/SiO₂-ZrO₂ to enhance the stability of the catalyst. It allows suggesting that the potential Cu-based catalysts for hydrogen production through ethanol dehydrogenation and dehydrogenative coupling are Cu/SiO₂-AE doped ZrO₂ and Ni

metal, where SiO₂ support can be MFI structure or mesoporous structure. However, SiO₂ support exhibits a low ethanol consumption rate in comparison with ZrO₂, MgAlO_x, C/SiC, and SiO₂/SiC supports (Table 1). Among them, C/SiC and SiO₂/SiC support are suitable for acetaldehyde production, while ZrO₂ is proper for ethyl acetate production (Fig. 12). Porous MgAlO_x is also a good support to produce hydrogen, acetaldehyde, and ethyl acetate (entry 15) [82].

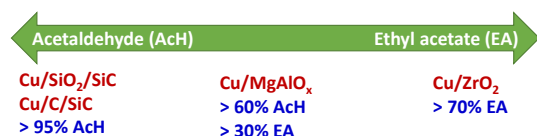


Fig. 12 – The scheme of potentials catalysts with a high ethanol consumption rate for acetaldehyde and ethyl acetate production.

However, the stability of Cu/MgAlO_x needs to improve to achieve the industrial requirement. That is a big gap and challenge between the research level and industry level. Also, the knowledge gap between absolute ethanol conversion and the rates of product formation (or ethanol consumption) should minimize to obtain a potential catalyst to develop for the industrial level. Additionally, the challenge is to create a catalyst with a high catalytic activity, stability, cheap price and easy production process. Cu/MgAlO_x , Cu/ZrO_2 and $\text{Cu/SiO}_2/\text{SiC}$ are potential candidates to develop to achieve industrial level.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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